Teoria das Comunicações

Princípios da Modulação em Amplitude (AM)

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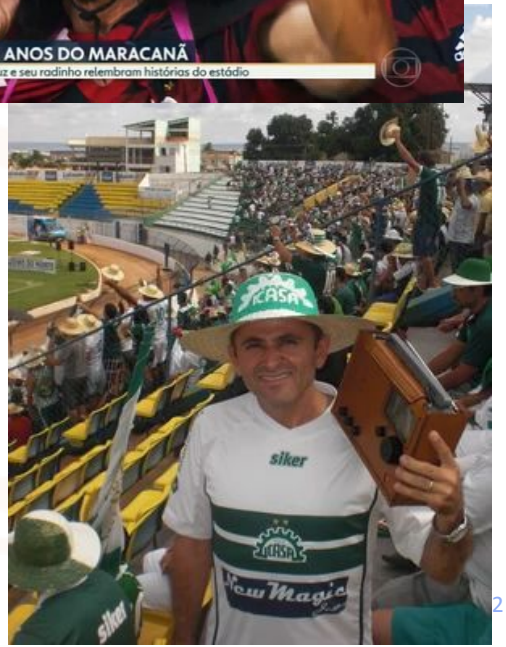
FCT/UFPA

Porque o rádio AM/FM continua a ser usado por torcedores?

- São mais rápidos?

- São mais baratos?

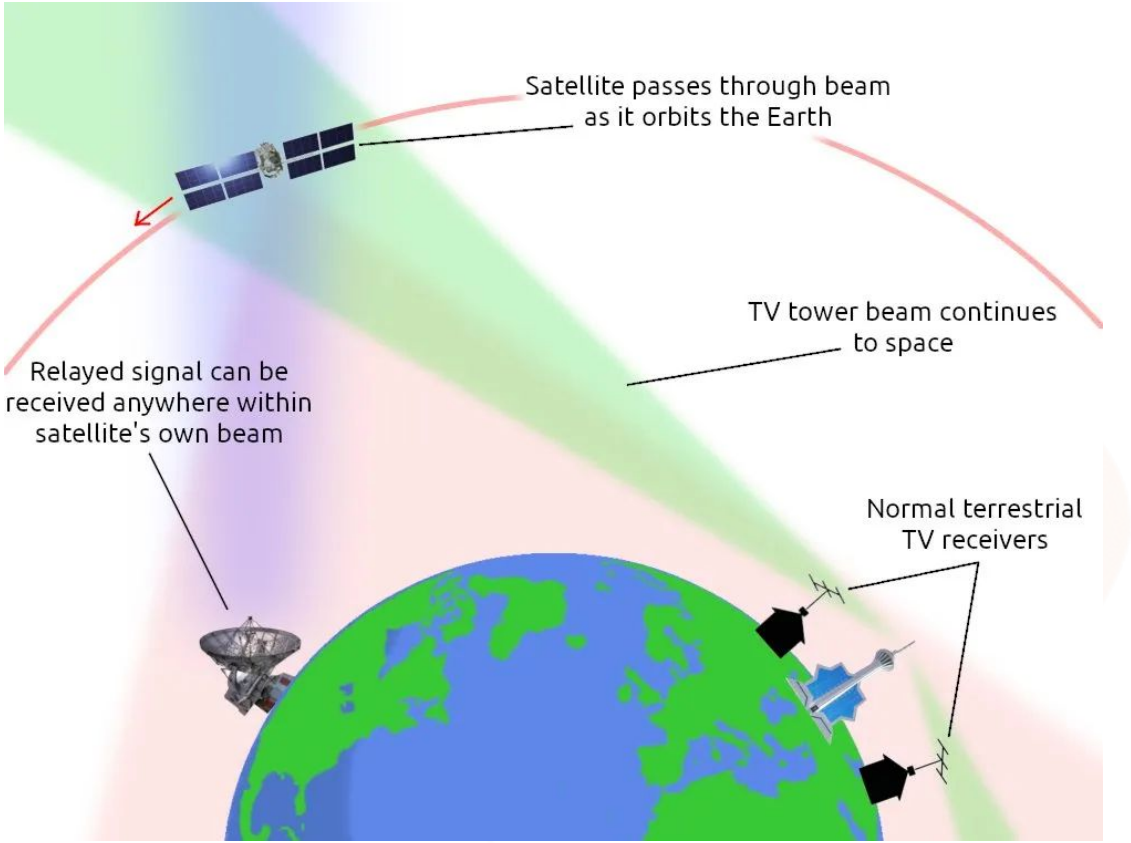
- São mais estáveis?



São mais rápidos? Depende em relação a que

- A velocidade da Luz é 300000 km/s idendependente da tecnologia de RF.

- O sinal da TV por satélite demora mais porque o transmissor está mais longe



A transmissão do sinal de TV “normal” é mais lento do que o de rádio?

- Os sinais de TV ou Rádio normalmente são transmitidos por antenas terrestres e navegam a mesma distância

- Porém o processamento do sinal de TV é mais complexo, incluindo codecs de vídeo, uso de IA para upscaling podem fazer com que a imagem demore mais para “chegar” ao telespectador do que o áudio do rádio.

Porque os rádios AM são baratos?

Os circuitos de um receptor de rádio AM são simples e de fácil fabricação

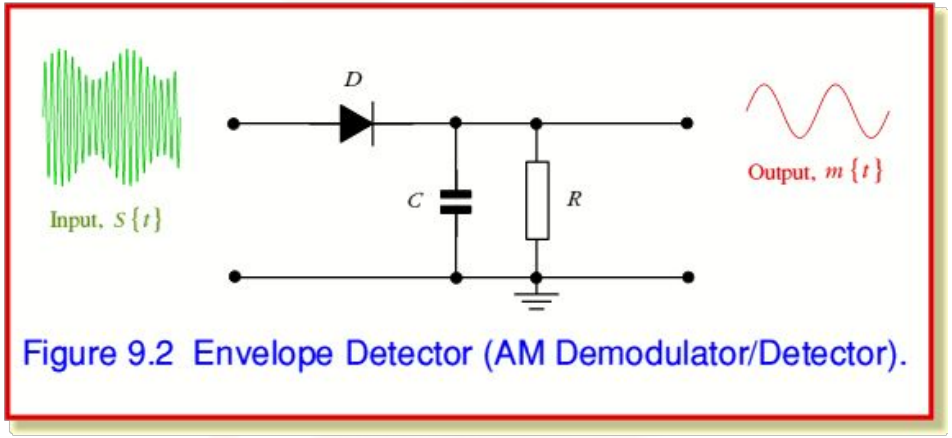
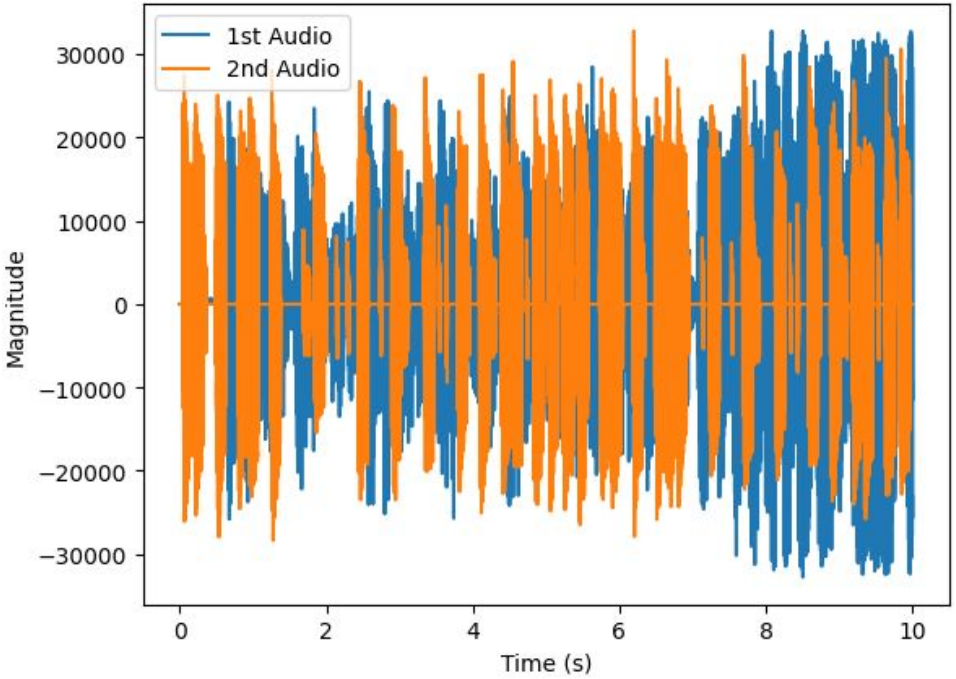
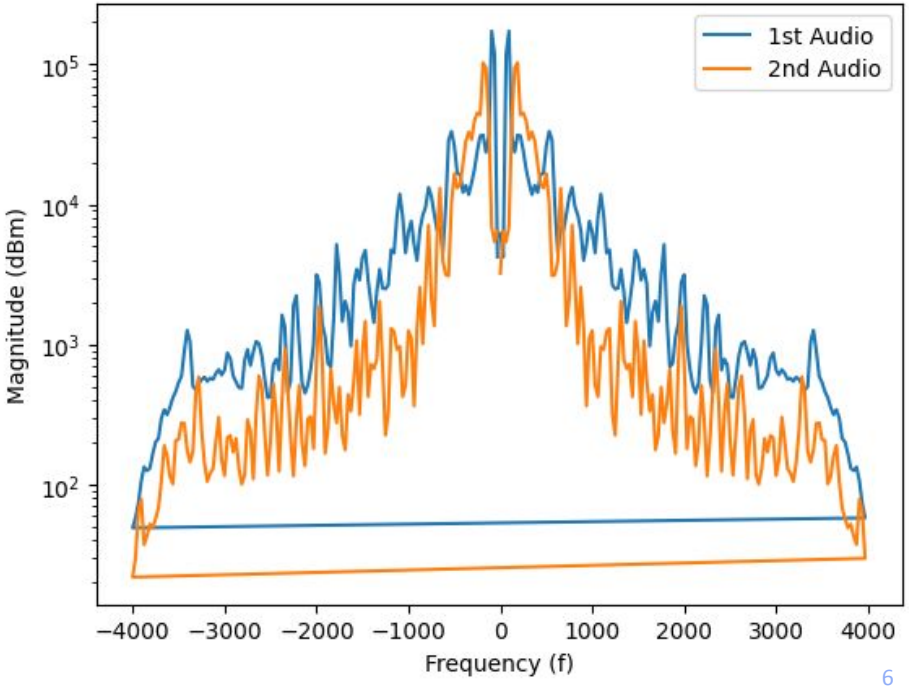


Figure 9.2 Envelope Detector (AM Demodulator/Detector).

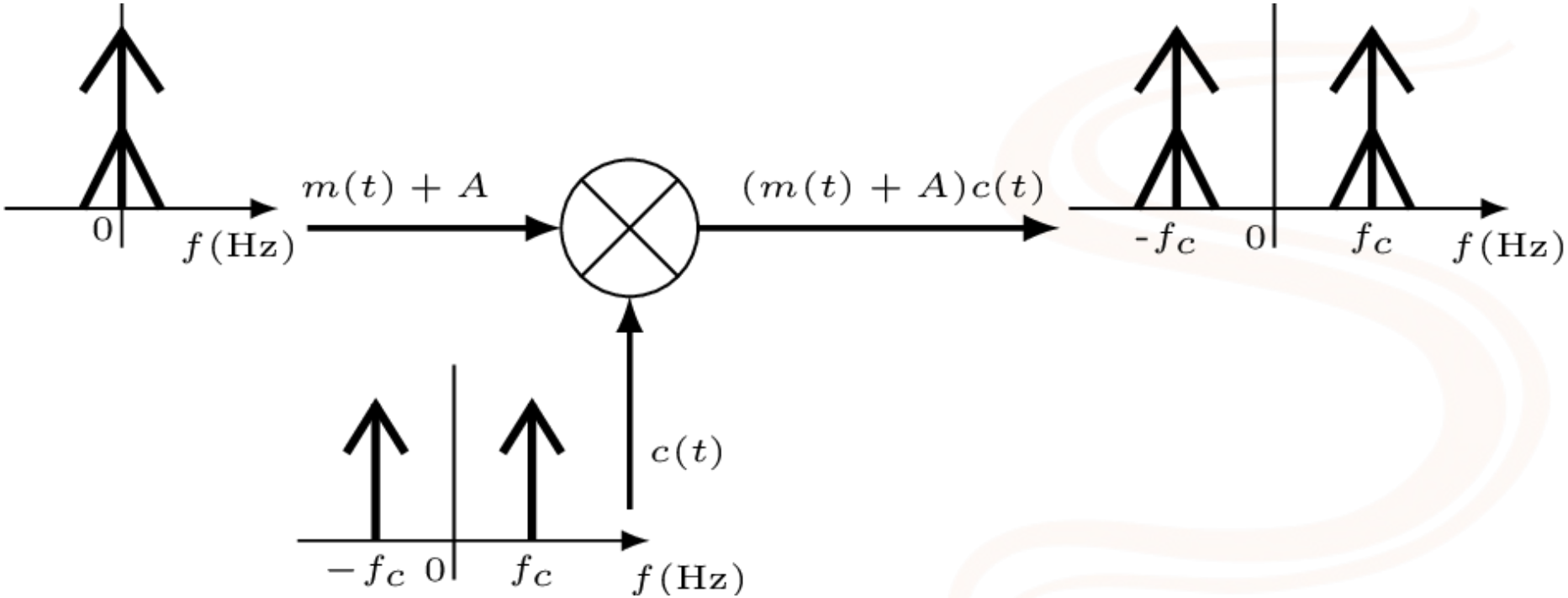
Enviando um áudio por transmissão AM

Qual a banda e a frequência máxima desse sinais? A informação está na amplitude do sinal

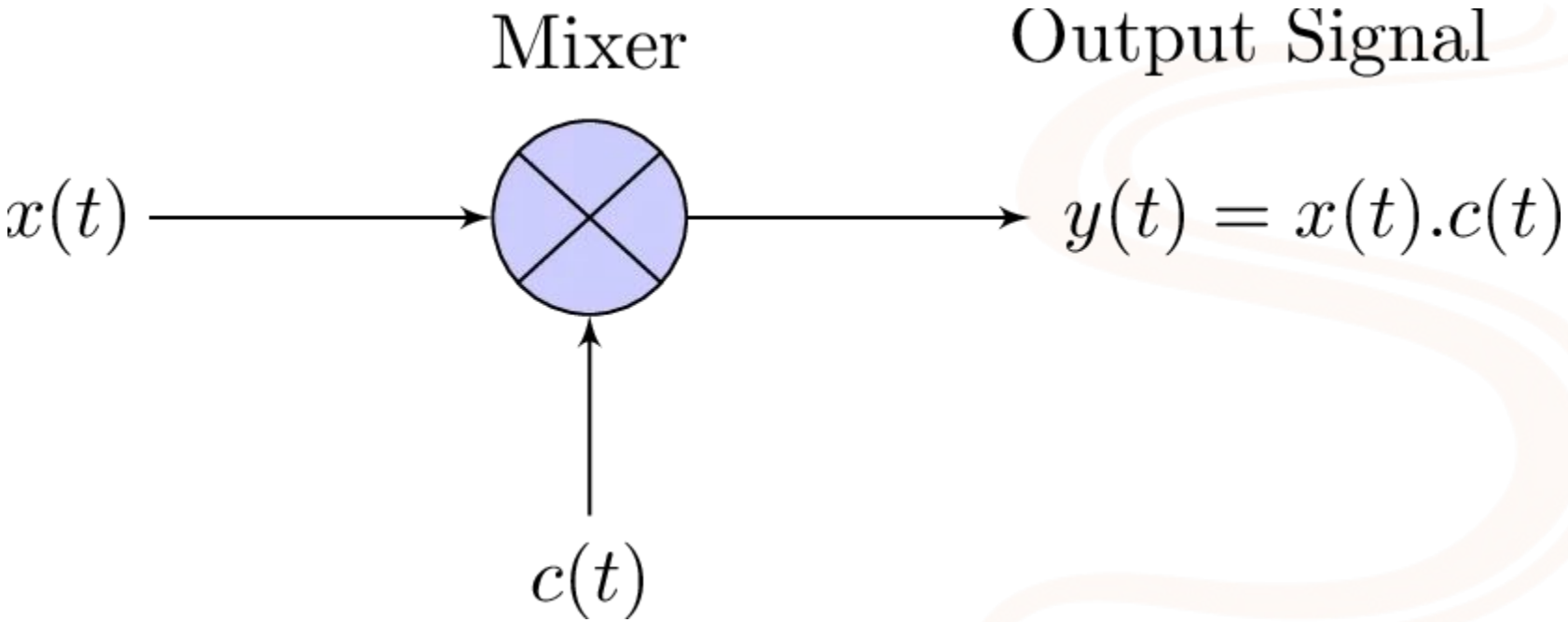




Como funciona um transmissor AM?



Usando mixers para transportar o nosso sinal para frequências maiores



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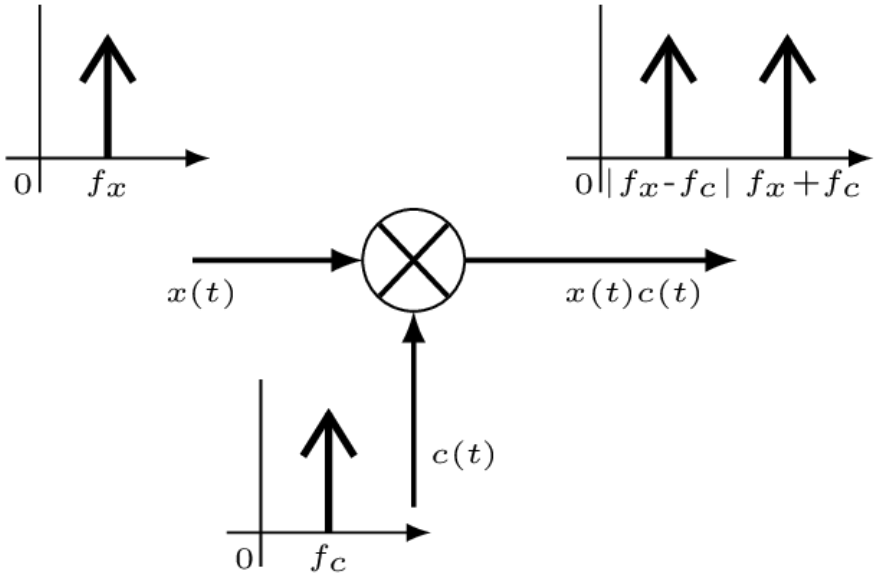
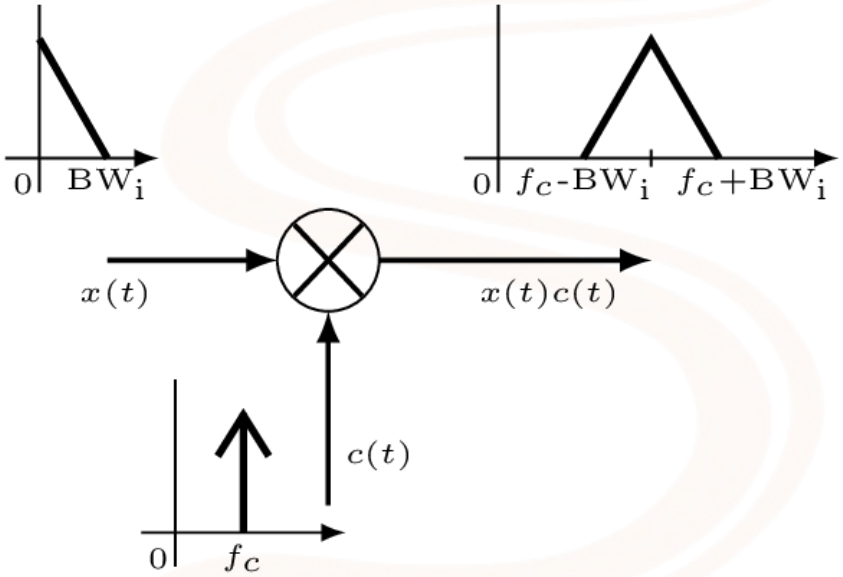
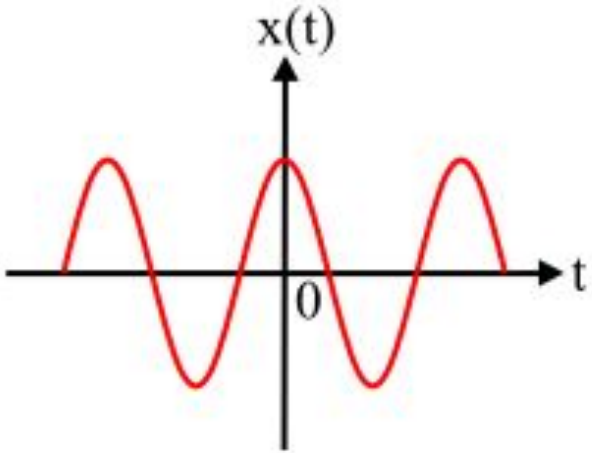


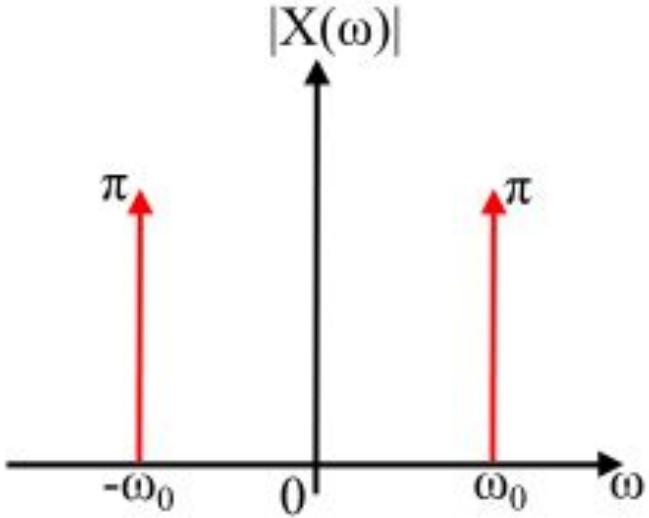


Relembrando a transformada de Fourier do Coseno

Assuma um cosseno com frequência w_0



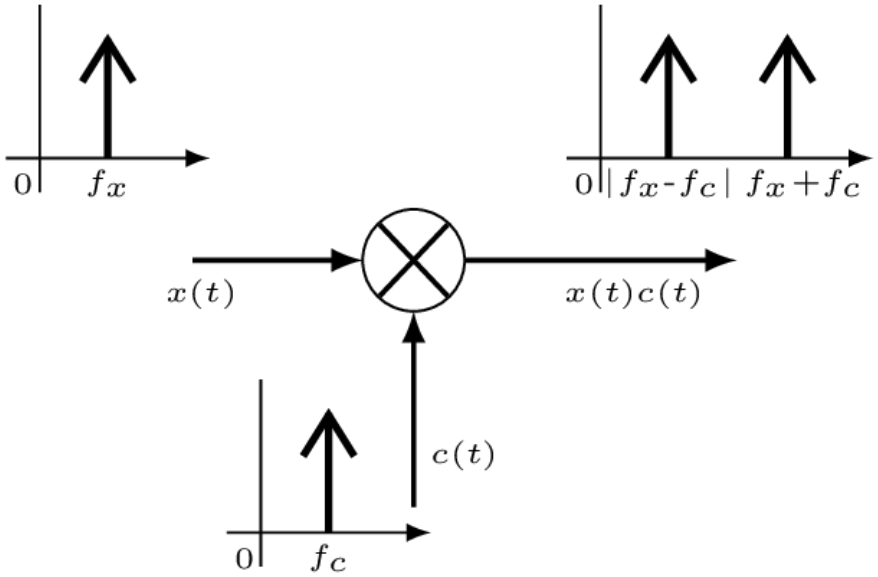
Cosine Wave

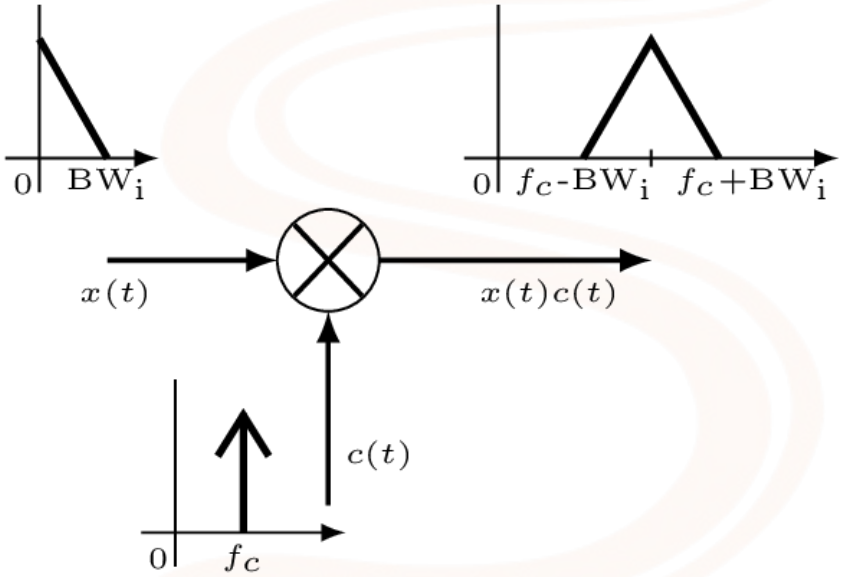


Magnitude Spectrum

Figure-2

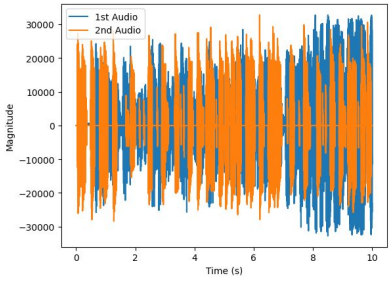
O sinal que é usado para transportar para outras frequências é normalmente um cosseno

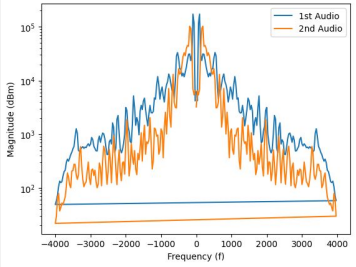




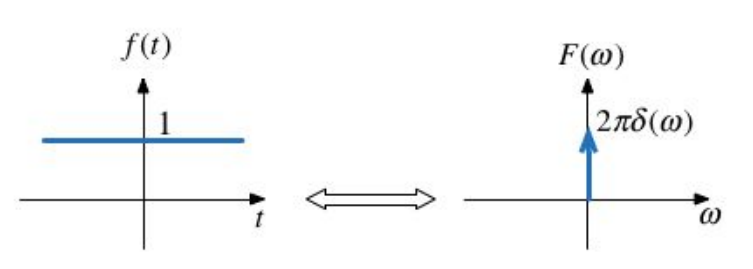
Voltando para o fluxo do transmissor AM

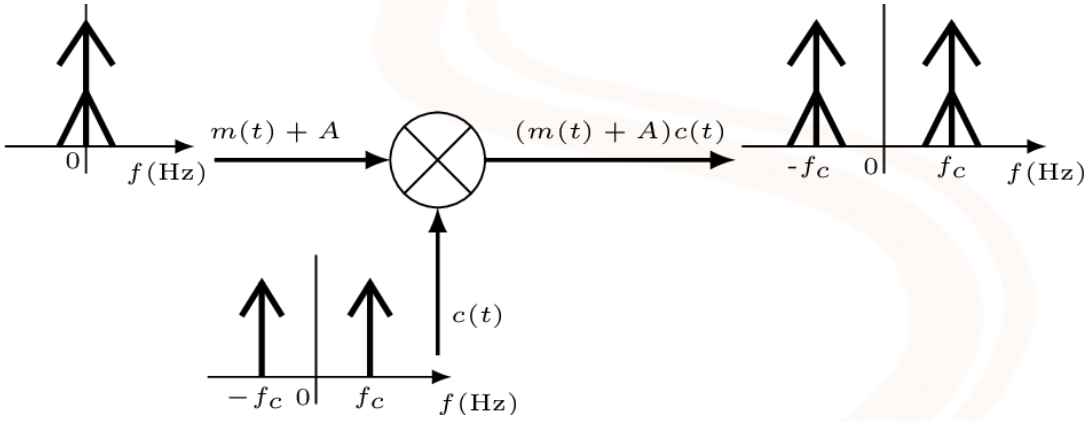
Áudio = $m(t)$





Portadora

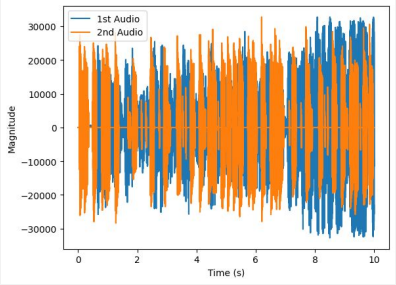


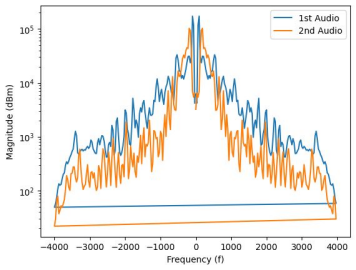


Transportando sinais para frequências mais altas no GNU Radio

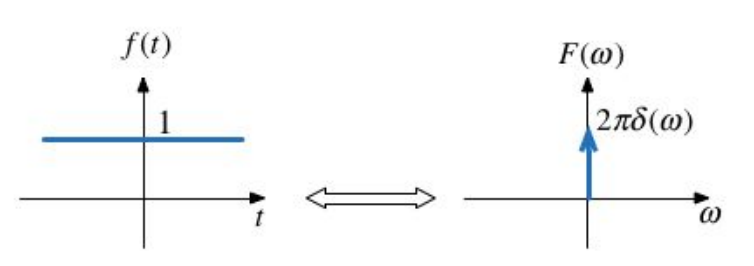
Porquê somar o sinal de áudio com uma constante?

Áudio = $m(t)$

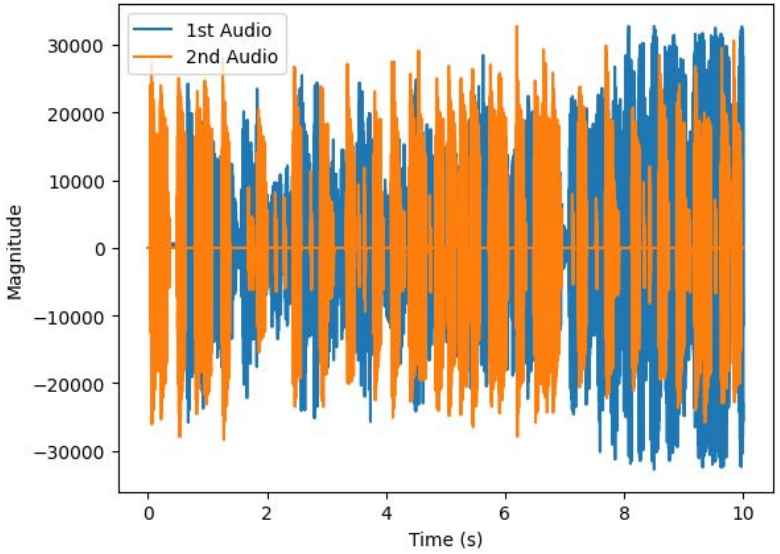


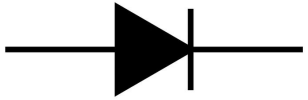


Portadora

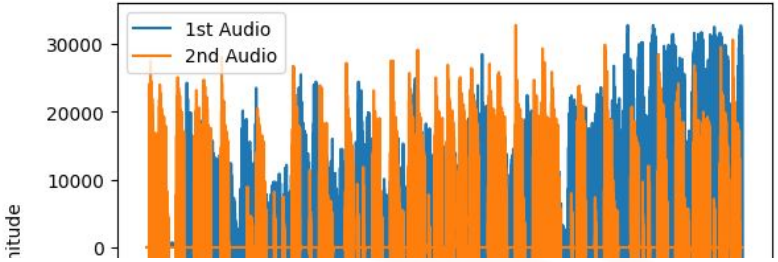


O sinal de áudio tem valores positivos e negativos. O diodo não lida bem com isso





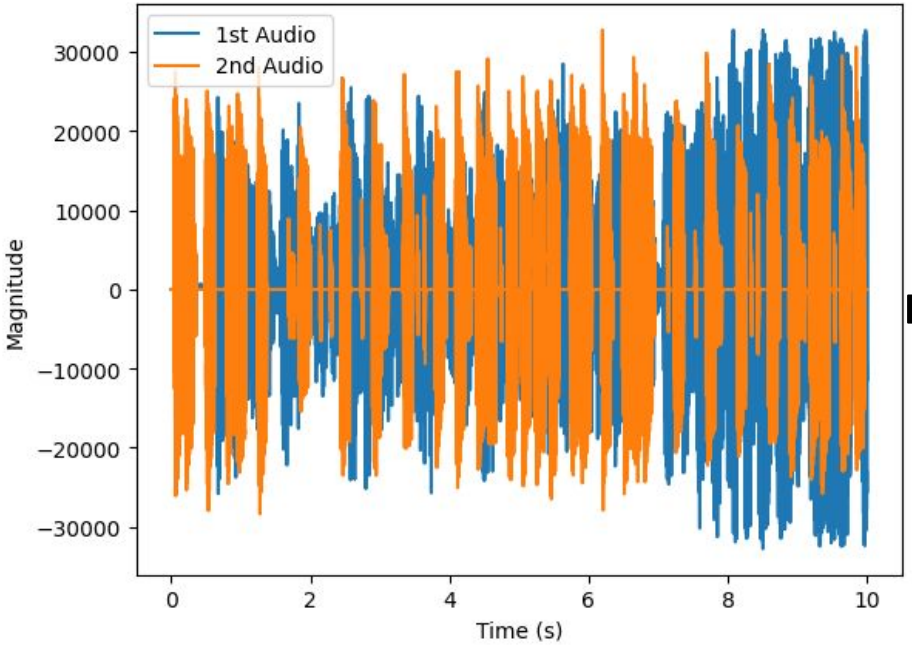
Diode

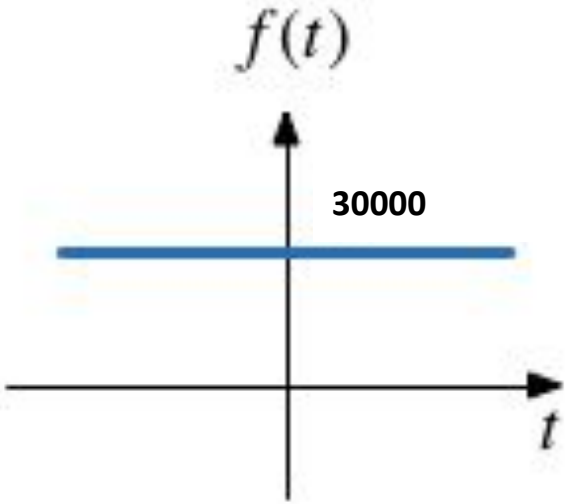


Apenas os valores nulos ou positivos restaram

* Processo simplificado

Somamos uma constante para não termos valor negativos no áudio





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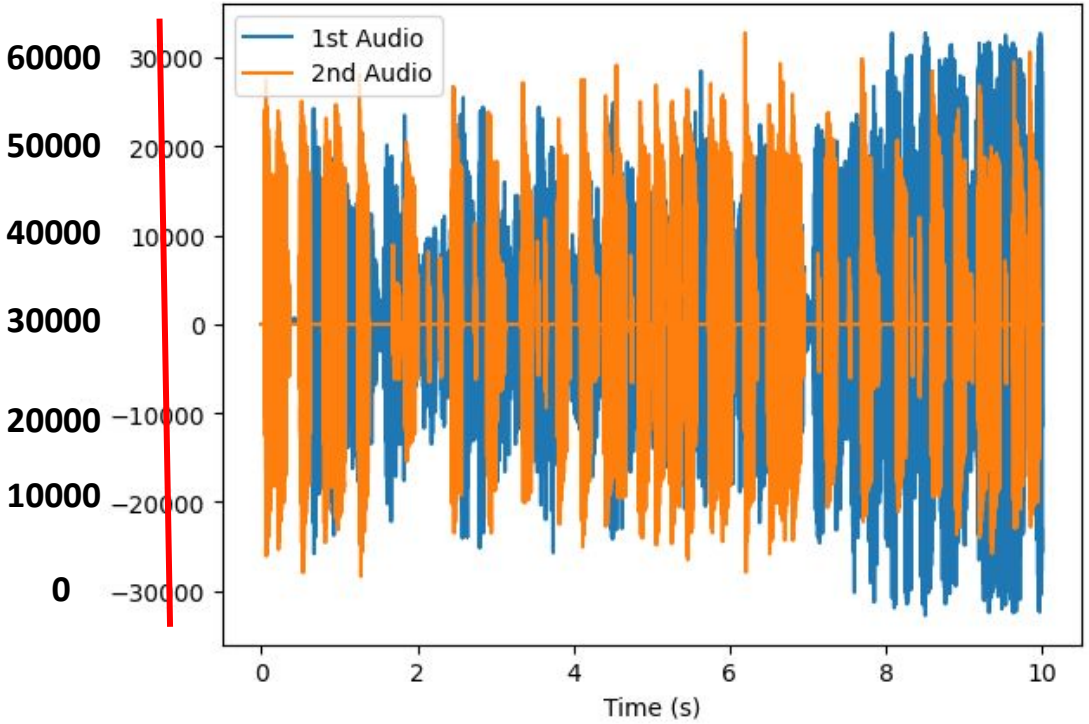
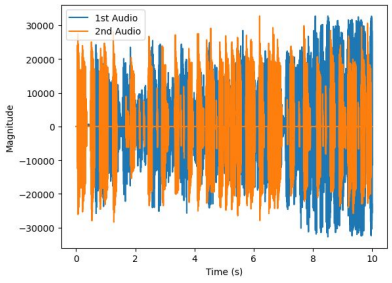
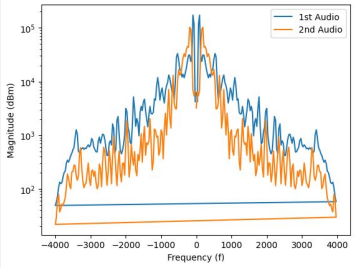


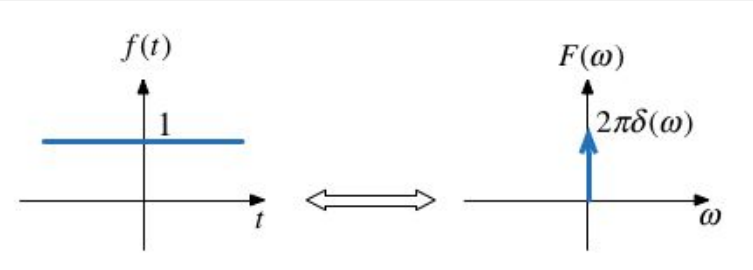
Com isso temos o nosso transmissor AM transmitindo em f_c

Áudio = $m(t)$

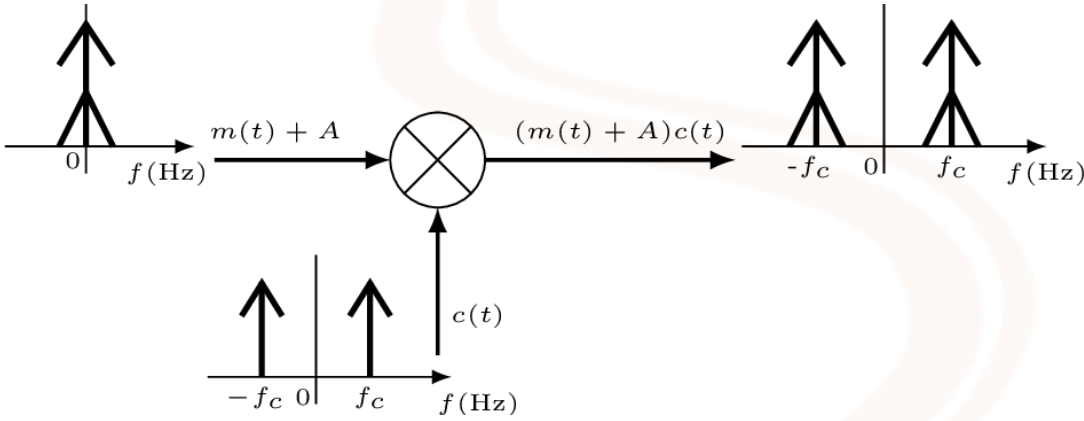




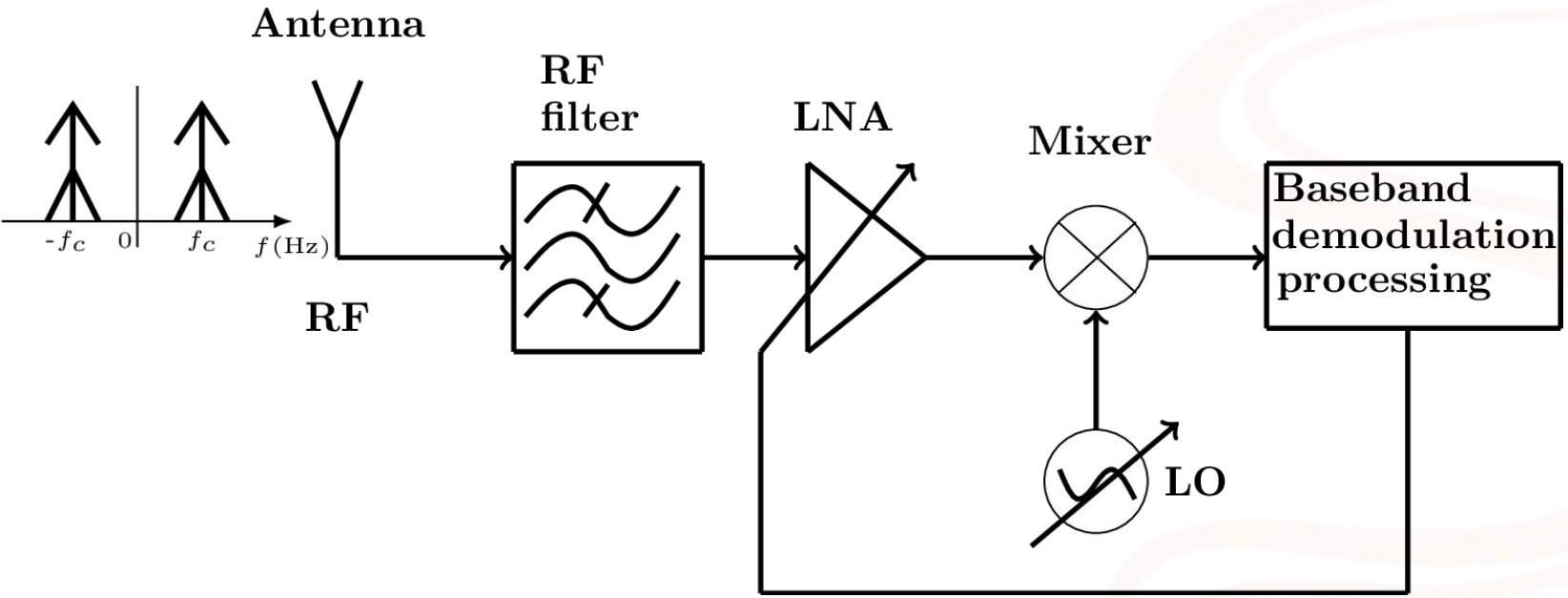
Portadora



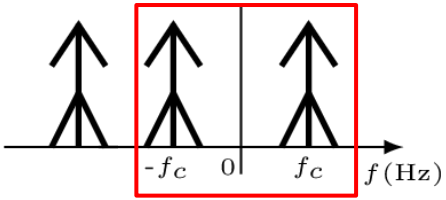
$$s(t) = (m(t) + A)\cos(2\pi f_c t + \phi)$$



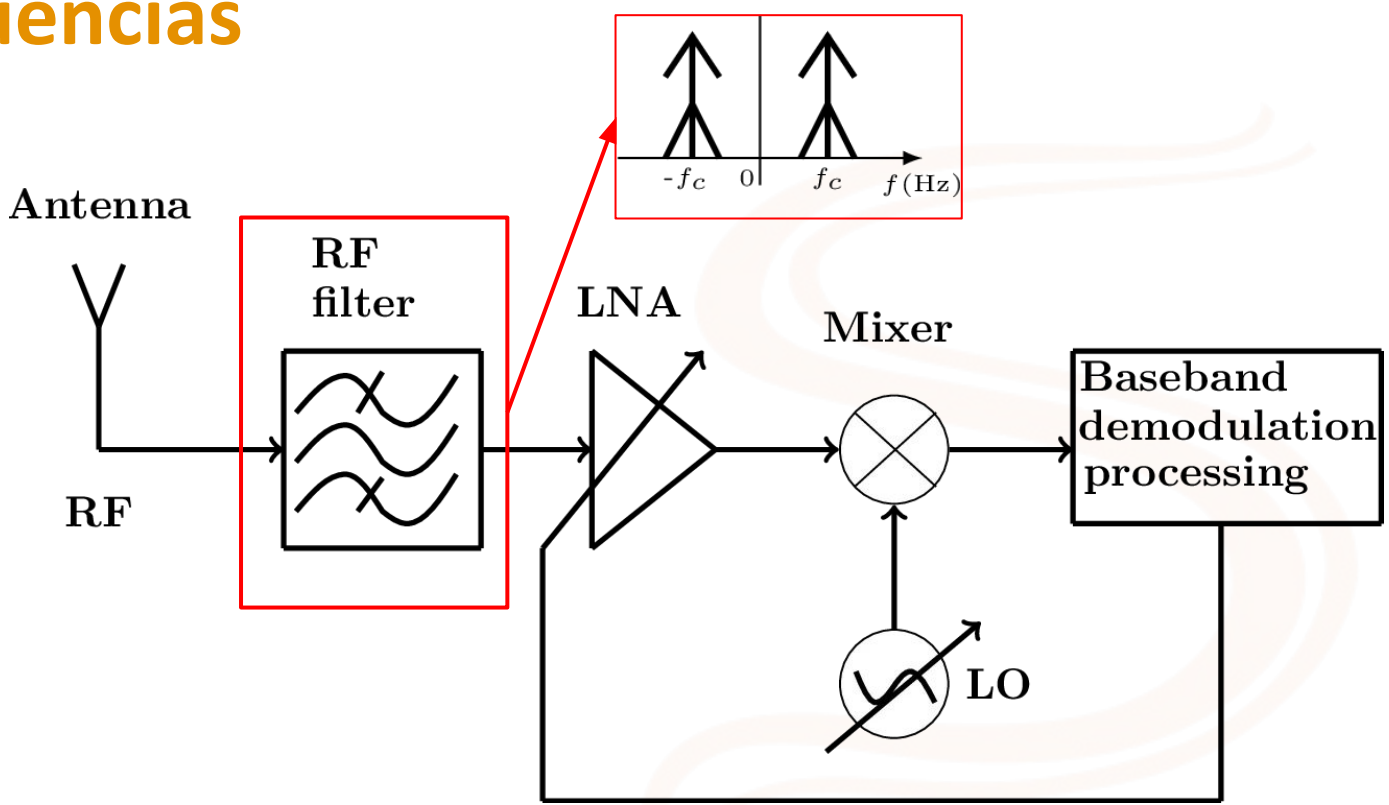
O receptor recebe o sinal em banda passante e precisa recuperar o sinal (áudio original)



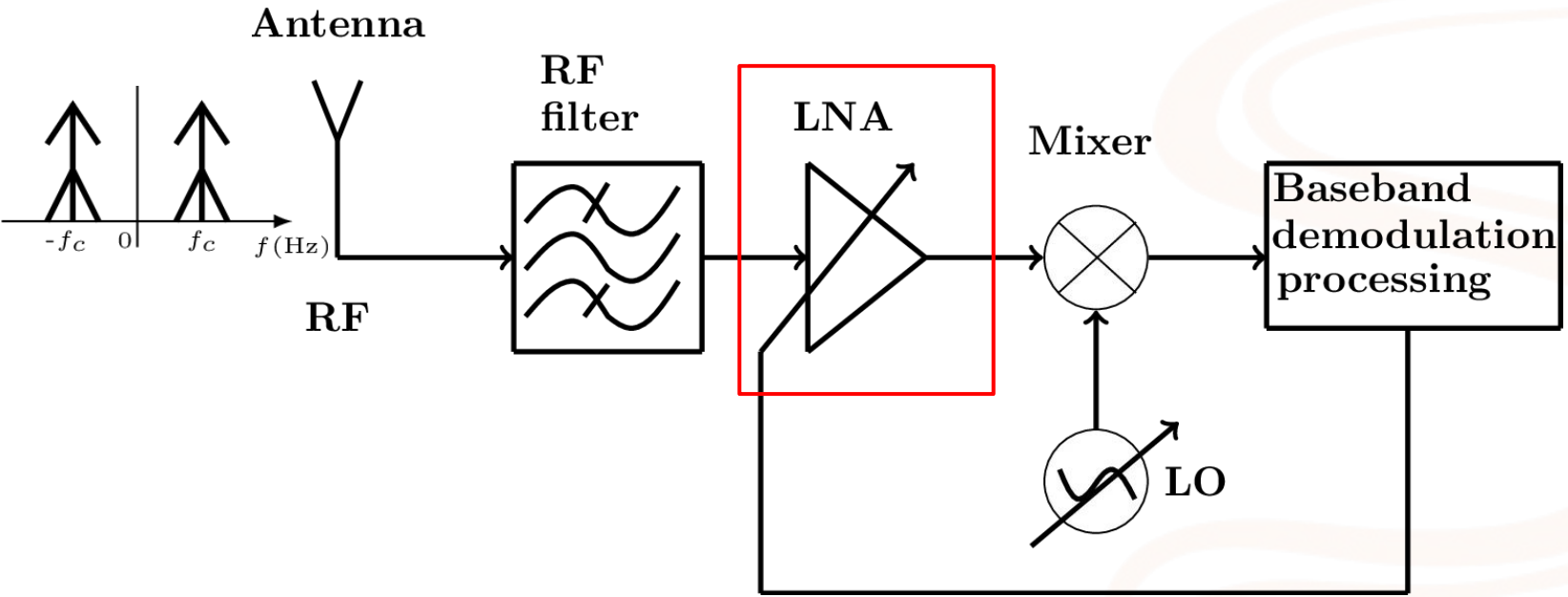
Filtramos o sinal para eliminar sinais em outras frequências



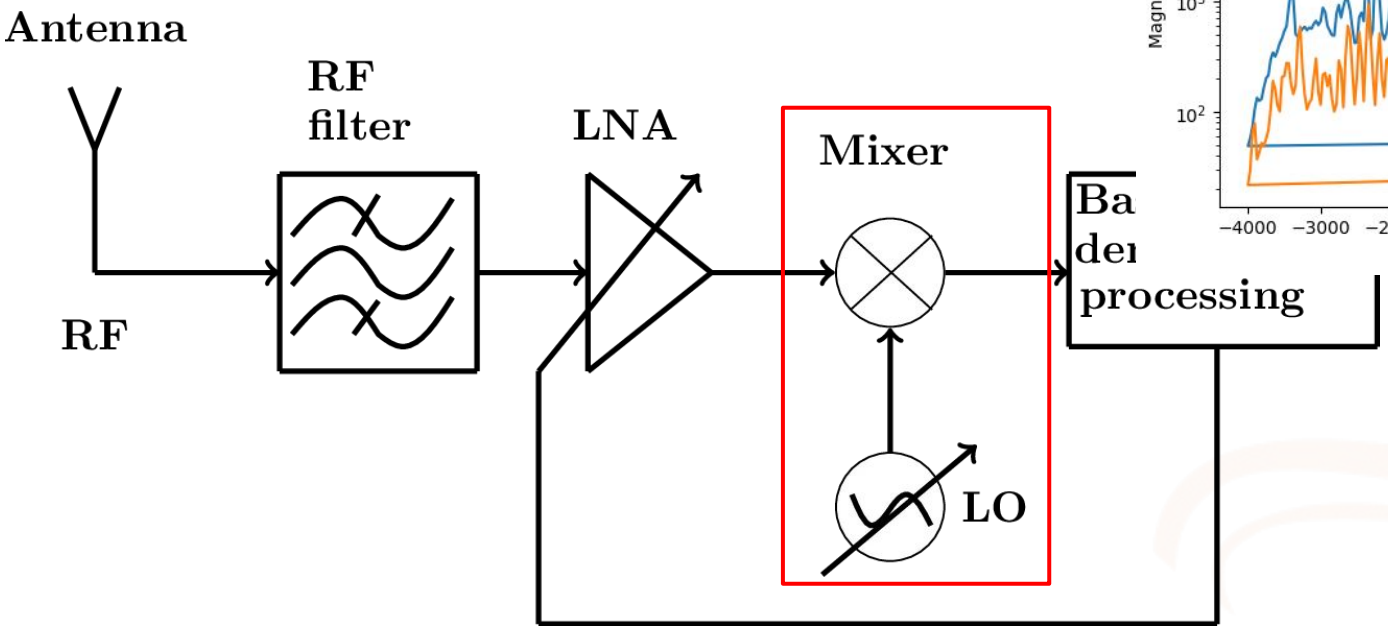
Sinal que queremos

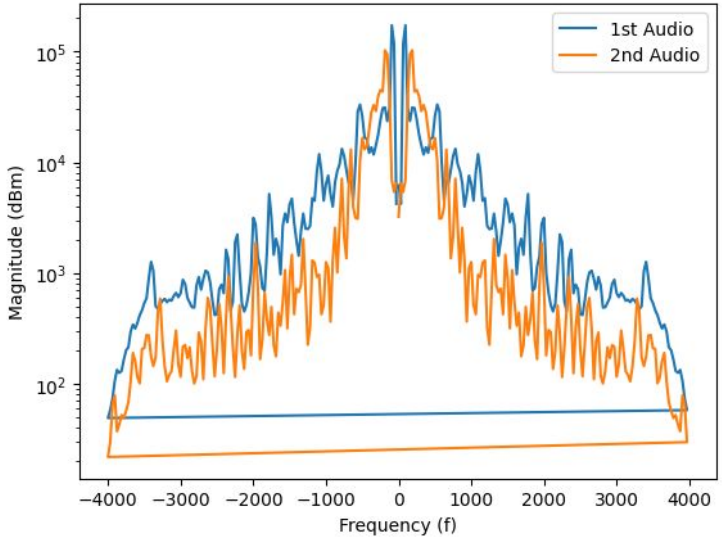


Low-noise amplifier: Aumentando a potência do sinal recebido sem aumentar tanto o ruído



Transportando o sinal de volta para banda-base





Transmissor e receptor de AM simples

Leituras recomendadas

- Transmissão AM
  - Livro com capítulo sobre transmissão AM https://ai6g.org/books/telecom/PrinciplesofCommunicationswithFocusonAM.html#principles-of-communications-with-focus-on-am
  - Código em Python implementando uma transmissão AM https://github.com/cleversonahum/digital_transmission_codes/blob/main/codes/am_transmission.ipynb
  - Capítulo 4 - Modulações e Demodulações em Amplitude - Livro Sistemas de Comunicações Analógicos e Digitais Modernos (4a Edição) - B. P. Lathi